

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1356

Cushaw 138 kV
December 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1356. The Interconnection Customer proposes a 185.9 MW wind facility interconnecting at the Cushaw 138 kV substation in the Virginia area. The request was received on December 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 505 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,527,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RUSENS / BOONSBORO 138 kV	BEAR / CLOVER 345 kV	132.3	388	24.2
MONEL SUBSTATION / CANDELARIA 138 kV line	BEAR / CLOVER 345 kV	121.2	443	5.7
CUSHAW / WHITE PINE 138 kV	CRANDALL SUBSTATION / ASPEN 345 kV	118.4	200	55.1
DEVIL'S HILL / WHITE PINE 138 kV	ASPEN / CRANDALL 345 kV	112.4	208	55.1
ISLA GRANDE GIS 11191120 / SNOWDEN 138 kV	CRANDALL / ASPEN 345 kV	108.5	211	55.1
ISLA GANDE GIS 11191120 / DEVIL'S HILL 138 kV	CRANDALL / ASPEN 345 kV	108.3	211	55.1
CARMEL 146TH ST / OAK CREEK 138 kV	REUSENS 345/1kV Auto	106.0	200	5.0
CEIBA BAJA 7012 / CARMEL 146TH ST 138 kV	REUSENS 345 kV/1 kV Transformer	105.6	204	5.0
BEAR / MECKLENBURG POWER STATION 345 kV	BEAR / CLOVER 345 kV	103.9	2,827	13.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RUSENS / BOONSBORO 138 kV	Rebuild line with higher-capacity conductor	\$4,860,000	37% (\$1,812,000)
MONEL SUBSTATION / CANDELARIA 138 kV line	Rebuild line with higher-capacity conductor	\$32,138,000	17% (\$5,335,000)
CUSHAW / WHITE PINE 138 kV	Rebuild line with higher-capacity conductor	\$6,563,000	83% (\$5,469,000)
DEVIL'S HILL / WHITE PINE 138 kV	Rebuild line with higher-capacity conductor	\$5,513,000	83% (\$4,594,000)
ISLA GRANDE GIS 11191120 / SNOWDEN 138 kV	Reconductor line	\$1,417,000	83% (\$1,181,000)
ISLA GANDE GIS 11191120 / DEVIL'S HILL 138 kV	Reconductor line	\$2,172,000	83% (\$1,810,000)
CARMEL 146TH ST / OAK CRK 138 kV	Reconductor line	\$2,192,000	29% (\$631,000)
CEIBA BAJA 7012 / CARMEL 146TH ST 138 kV	Reconductor line	\$2,618,000	29% (\$753,000)
BEAR / MECKLENBURG POWER STA 345 kV	Reconductor line	\$51,062,000	2% (\$942,000)
	Total allocated to project		\$22,527,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.